

Customer Shopping Behavior SQL Analysis Report

Customer-level purchasing, discount, subscription and product analysis

Project	Customer Shopping Behavior Analysis
Tools	Python • Pandas • PostgreSQL / SQL • Power BI
Dataset size	3,900 customer purchase records
Dashboard	Customer Behavior Dashboard

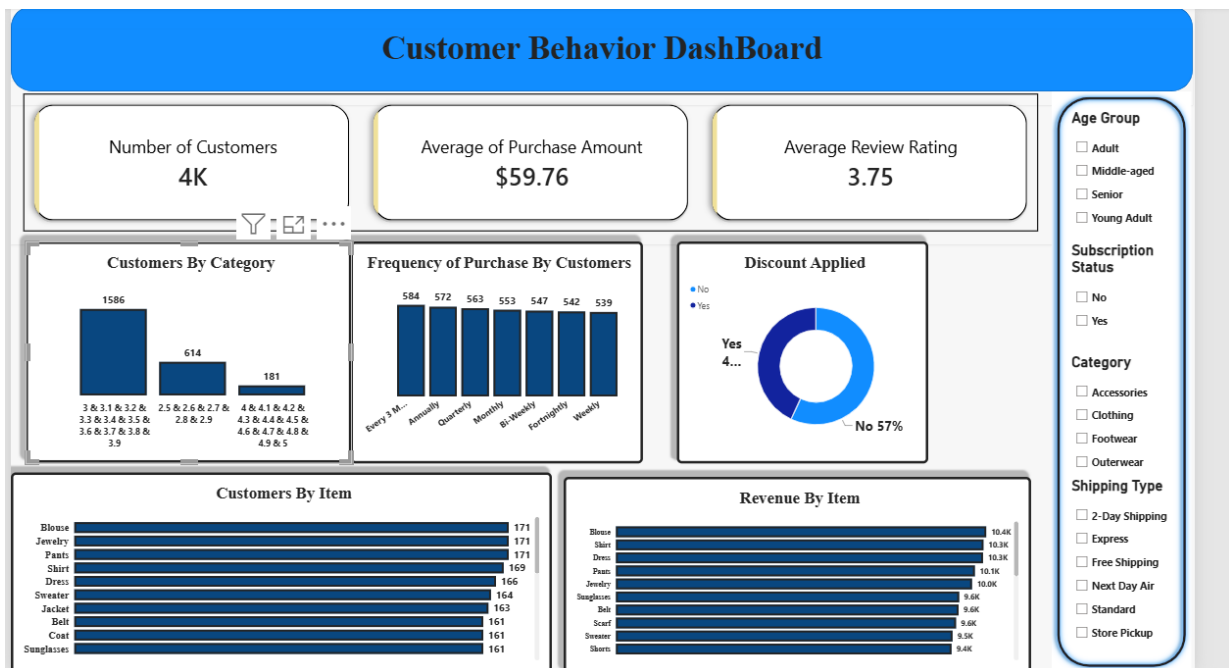
Prepared for GitHub project documentation

1. Project Overview

This project analyzes customer shopping behavior using Python for preparation and SQL for business-oriented analysis, with the results presented in a Power BI dashboard. The analysis focuses on customer counts, purchase value, review ratings, purchase frequency, discounts, subscriptions, product performance and age-group revenue.

The uploaded notebook shows a 3,900-row dataset with 18 initial columns. The workflow includes data inspection, missing-value treatment, column standardization, age segmentation, purchase-frequency conversion and loading the prepared data into a relational database.

Dashboard Snapshot



Dashboard values visible in the supplied screenshot: 4K customers, average purchase amount \$59.76, average review rating 3.75, and a discount-status split showing 57% without a discount. The dashboard also includes category, purchase-frequency, item-count, revenue-by-item and slicer views.

2. Data Preparation

Step	Implementation
Data loading	Loaded customer_shopping_behavior.csv into a Pandas DataFrame.
Data inspection	Checked structure, summary statistics and missing values.
Missing values	Review Rating nulls were imputed using the median review rating within each product category.
Column standardization	Converted column names to lowercase snake_case and renamed purchase_amount_(usd) to purchase_amount.
Age segmentation	Created age_group using quartiles with Young Adult, Adult, Middle-aged and Senior labels.
Purchase frequency	Mapped purchase-frequency labels to approximate day intervals such as Weekly=7, Monthly=30 and Annually=365.
Redundant field	promo_code_used was removed after checking it against discount_applied.

Step	Implementation
Database loading	Prepared data was loaded into a customer table for SQL analysis; the notebook documents PostgreSQL, MySQL and SQL Server connection examples.

3. SQL Business Analysis

The following ten SQL questions form the analytical layer of the project. They cover revenue, discount behavior, product ratings, shipping, subscriptions, customer segmentation, category-level product ranking, repeat buying and age-group revenue.

Q1. Revenue by gender

```
SELECT gender,
       SUM(purchase_amount) AS revenue
FROM customer
GROUP BY gender;
```

Calculates total purchase revenue separately for male and female customers.

Q2. Discount users spending above average

```
SELECT customer_id, purchase_amount
FROM customer
WHERE discount_applied = 'Yes'
  AND purchase_amount >= (
    SELECT AVG(purchase_amount)
    FROM customer
  );
```

Uses a scalar subquery to compare each discounted purchase against the overall average purchase amount.

Q3. Top 5 products by average review rating

```
SELECT item_purchased,
       ROUND(AVG(review_rating::numeric), 2) AS "Average Product Rating"
FROM customer
GROUP BY item_purchased
ORDER BY AVG(review_rating) DESC
LIMIT 5;
```

Aggregates review ratings at product level and returns the five highest average ratings.

Q4. Standard vs Express shipping

```
SELECT shipping_type,
       ROUND(AVG(purchase_amount), 2) AS avg_purchase_amount
FROM customer
WHERE shipping_type IN ('Standard', 'Express')
GROUP BY shipping_type;
```

Compares average purchase amounts for the two specified shipping methods.

Q5. Subscriber vs non-subscriber spending

```
SELECT subscription_status,
       COUNT(customer_id) AS total_customers,
       ROUND(AVG(purchase_amount), 2) AS avg_spend,
       ROUND(SUM(purchase_amount), 2) AS total_revenue
FROM customer
GROUP BY subscription_status
ORDER BY total_revenue, avg_spend DESC;
```

Provides customer count, average spend and total revenue for subscription groups.

Q6. Products with the highest discount-purchase rate

```
SELECT item_purchased,
       ROUND(
         100.0 * SUM(
           CASE WHEN discount_applied = 'Yes' THEN 1 ELSE 0 END
         ) / COUNT(*), 2
       ) AS discount_rate
FROM customer
GROUP BY item_purchased
ORDER BY discount_rate DESC
LIMIT 5;
```

Measures the percentage of purchases receiving a discount for each product and returns the top five.

Q7. Customer segmentation

```

WITH customer_type AS (
  SELECT customer_id, previous_purchases,
    CASE
      WHEN previous_purchases = 1 THEN 'New'
      WHEN previous_purchases BETWEEN 2 AND 10 THEN 'Returning'
      ELSE 'Loyal'
    END AS customer_segment
  FROM customer
)
SELECT customer_segment,
  COUNT(*) AS "Number of Customers"
FROM customer_type
GROUP BY customer_segment;

```

Creates New, Returning and Loyal segments from previous-purchase counts, then counts customers in each segment.

Q8. Top 3 products within each category

```

WITH item_counts AS (
  SELECT category,
    item_purchased,
    COUNT(customer_id) AS total_orders,
    ROW_NUMBER() OVER (
      PARTITION BY category
      ORDER BY COUNT(customer_id) DESC
    ) AS item_rank
  FROM customer
  GROUP BY category, item_purchased
)
SELECT item_rank, category, item_purchased, total_orders
FROM item_counts
WHERE item_rank <= 3;

```

Uses ROW_NUMBER with PARTITION BY category to rank products independently inside every category.

Q9. Repeat buyers and subscription status

```

SELECT subscription_status,
  COUNT(customer_id) AS repeat_buyers
FROM customer
WHERE previous_purchases > 5
GROUP BY subscription_status;

```

Filters customers with more than five previous purchases and compares their subscription-status counts.

Q10. Revenue by age group

```

SELECT age_group,
  SUM(purchase_amount) AS total_revenue
FROM customer
GROUP BY age_group
ORDER BY total_revenue DESC;

```

Aggregates total purchase revenue across the age groups created during data preparation.

4. Dashboard-Based Observations

Metric / View	Visible result / interpretation
Number of customers	Approximately 4K, consistent with the 3,900 records in the source dataset after dashboard rounding.
Average purchase amount	\$59.76 shown in the KPI card.
Average review rating	3.75 shown in the KPI card.
Discount applied	The donut chart shows 57% of purchases without a discount; the remaining share is approximately 43% with a discount.
Customer category view	The dashboard displays three aggregated category groups with visible counts of 1,586, 614 and 181.
Purchase frequency	The displayed bars show the highest count for the Every 3 Months group (584), followed by Annually (572) and Quarterly (563).
Customers by item	Blouse and Jewelry are shown at 171 customers each, followed by Pants at 171 in the supplied visual.
Revenue by item	Blouse is shown at approximately \$10.4K, followed by Shirt at approximately \$10.3K and Dress at approximately \$10.3K.

5. Technical Skills Demonstrated

- Python / Pandas: data loading, inspection, null handling, feature creation and column standardization.
- SQL: GROUP BY, aggregate functions, CASE expressions, scalar subqueries, CTEs, ROW_NUMBER and PARTITION BY.
- Relational databases: loading the prepared dataset into a customer table using SQLAlchemy.
- Power BI: KPI cards, bar charts, donut chart and interactive slicers for customer behavior analysis.
- Business analysis: revenue comparison, discount analysis, customer segmentation, product ranking and subscription analysis.

6. GitHub Repository Structure Recommendation

File	Purpose
Customer_Shopping_Behavior_Analysis.ipynb	Python data preparation and database-loading workflow.
customer_shopping_behavior_sql_analysis_report.pdf	This project report documenting the SQL analysis and dashboard.
dashboard.png	Exported image of the Power BI Customer Behavior Dashboard.
README.md	Project objective, dataset description, tools, analysis questions, dashboard screenshot and key findings.

Note: The SQL questions and dashboard observations in this report are based on the SQL supplied with this request, the supplied dashboard image, and the uploaded project notebook. Where an exact SQL result was not present in the supplied materials, the report documents the query and its purpose

rather than inventing a result.